BJT

$g_m = \frac{I_C}{V_T}$

$r_e = 1/g_m$

$r_\pi = (\beta+1) r_e$

Mosfet

$g_m = \frac{2 I_D}{V_{ov}}$

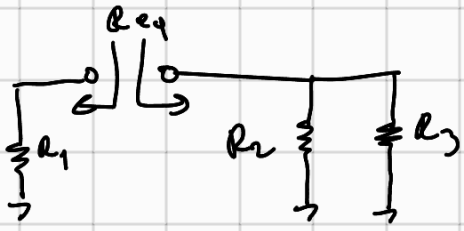
$r_o = \frac{|V_A|}{I_{C,D}}$

$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t)^2 (1 + \lambda V_{DS})$

$I_C = I_S e^{\frac{V_{BE}}{V_T}}$

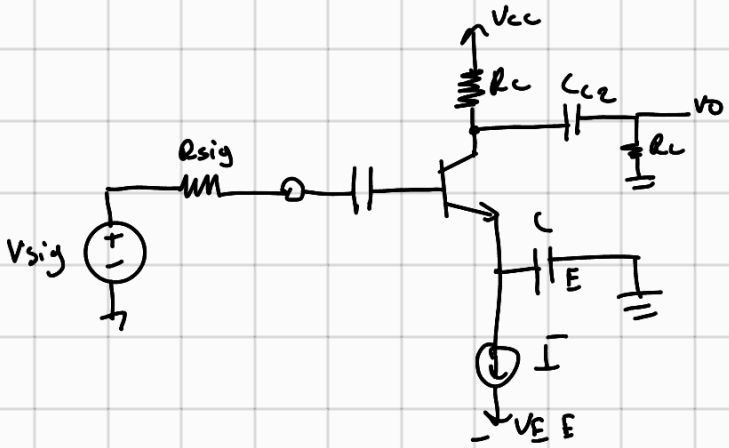
$f_L = \frac{1}{2\pi} \left(\underbrace{\frac{1}{R_1 C_1}}_{10} + \underbrace{\frac{1}{R_2 C_2}}_{50\ k} + \underbrace{\frac{1}{R_3 C_3}}_{100} \right)$

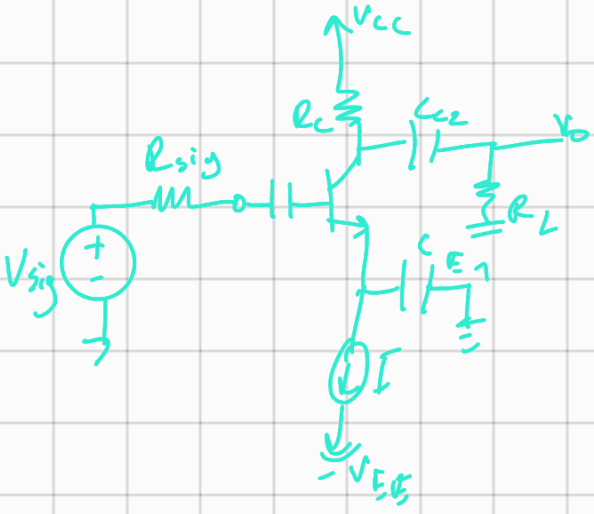
⇓ dominant

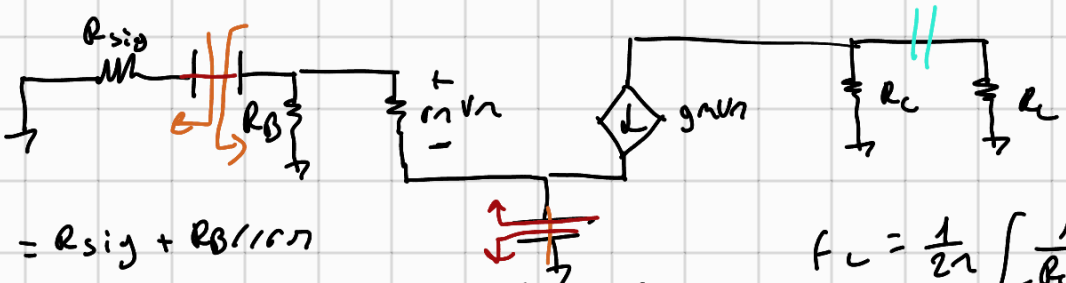


$R_{eq} = R_1 + R_2 // R_3$

The CE amplifier







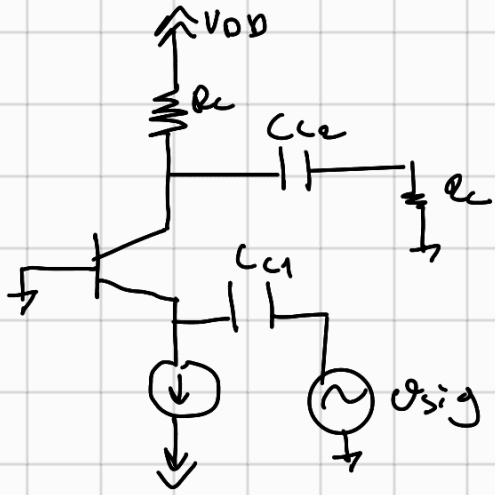
$R_{CC1} = R_{sig} + R_B // r_\pi$

$R_{E1} = \frac{R_{sig} // R_B + r_\pi}{\beta + 1} = r_e + \frac{R_{sig} // R_B}{\beta + 1}$

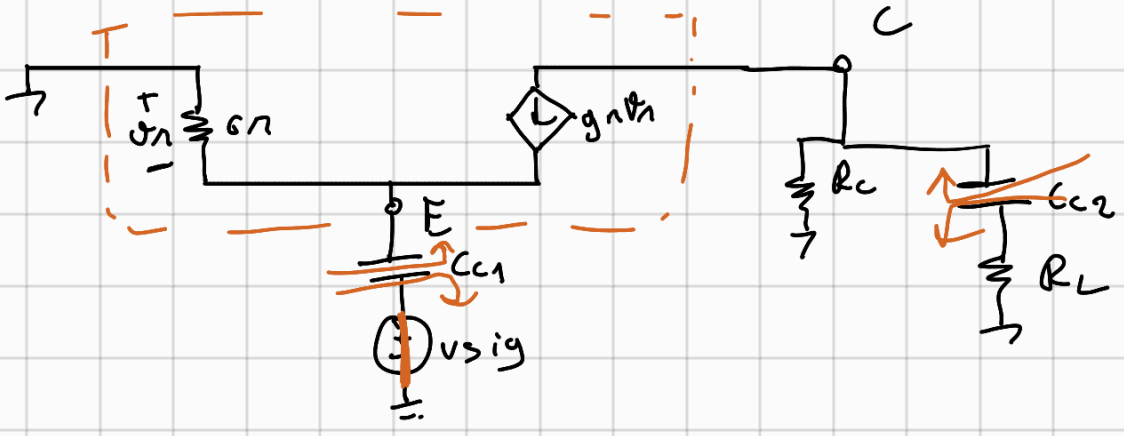
$f_L = \frac{1}{2\pi} \left[\frac{1}{R_{C1} C_{C1}} + \frac{1}{R_{C2} C_{C2}} + \frac{1}{R_E C_E} \right)$

$R_{out2} = R_C + R_L$

Sorry

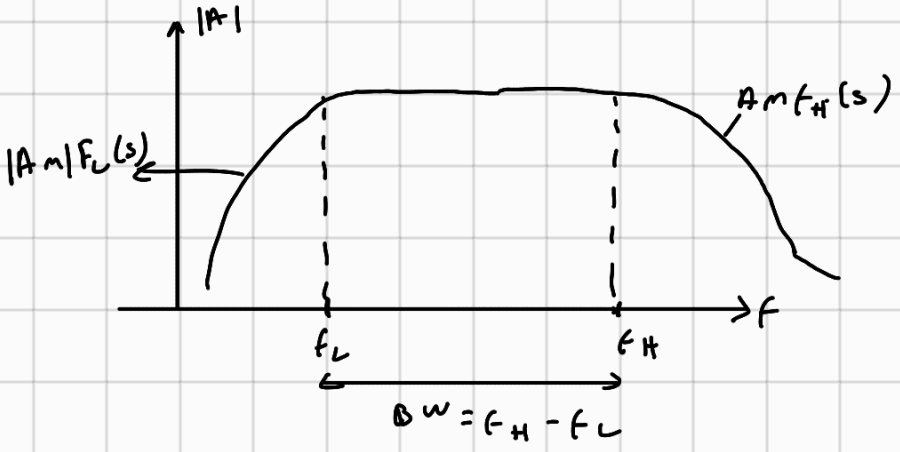


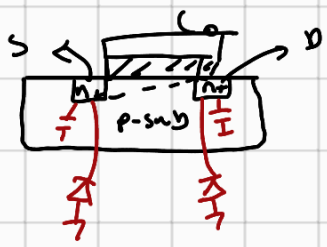
$f_L = ?$



from CC_1 $\quad R_{CC_1} \cong 6e$

from C_{c_2} $\quad R_{cc_2} = R_C + R_e$





Triode Region: $C_{gs} = C_{gd} = \frac{1}{2} WL C_{ox}$

Sat region:
$$\begin{cases} C_{gs} = \frac{2}{3} WL C_{ox} \\ C_{gd} = 0 \end{cases}$$

cut off region:
$$\begin{cases} C_{gs} = C_{gd} = 0 \\ C_{gb} = WL C_{ox} \end{cases}$$

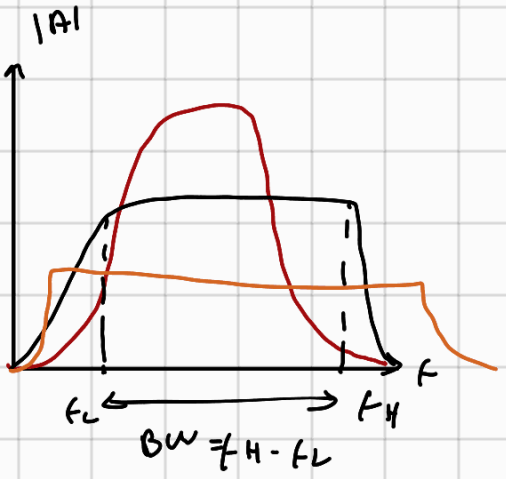
overlap capacitance: $C_{ov} = W L_{ov} C_{ox}$

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t)^2$$

$$|A_0| = g_m r_o$$

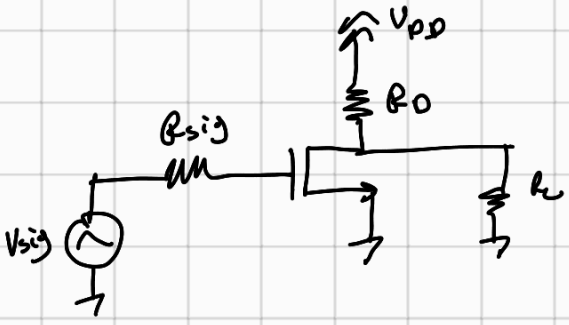
$$\downarrow \quad \frac{2I_D}{V_{ov}} \cdot \frac{V_A' L}{I_D} \quad \downarrow$$

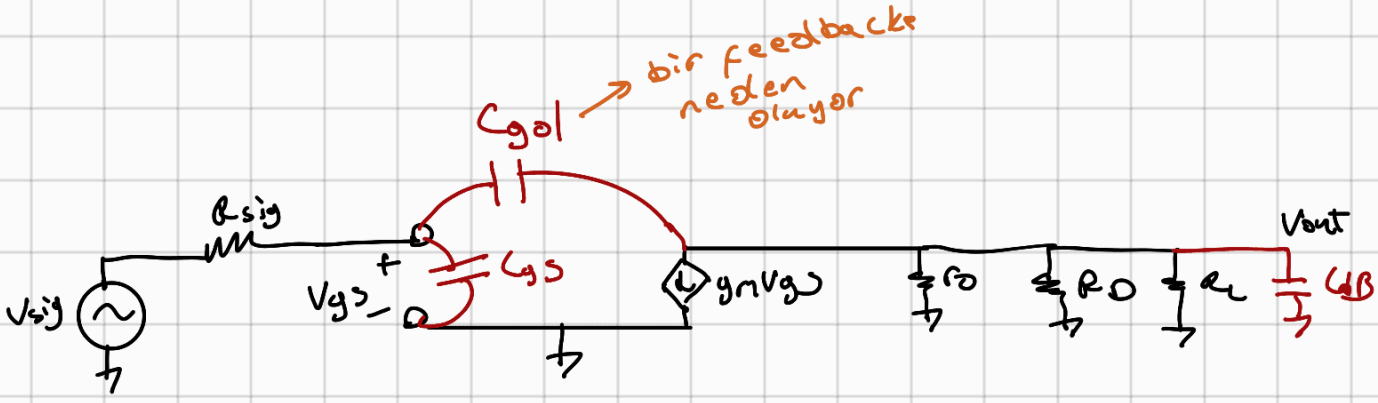
$$= \frac{2 \cdot V_A' L}{V_{ov}}$$



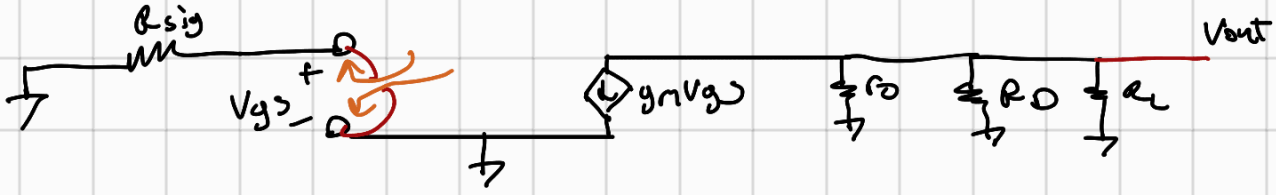
Source body capacitance: $C_{sb} = \frac{C_{sb0}}{\sqrt{1 + \frac{V_{SB}}{V_0}}}$

$$C_{db} = \frac{C_{db0}}{\sqrt{1 + \frac{V_{DB}}{V_0}}}$$

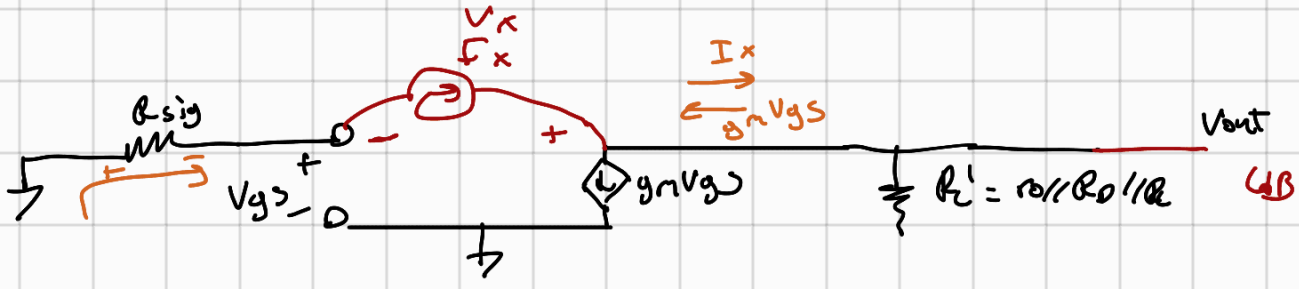




Analiz yaparken açıkdevre yapılırlar



$$R_{gs} = R_{sig} \,//$$



$\frac{V_x}{I_x} = ?$

$V_{out} = V_x + V_{gs} = (I_x - g_m V_{gs}) R_L'$

$V_{gs} = -R_{sig} I_x$

$V_x - R_{sig} I_x = (I_x + g_m R_{sig} I_x) R_L'$

$V_{out} = V_x + V_{gs}$

$= (I_x - g_m V_{gs}) R_L'$

$V_{gs} = -R_{sig} I_x$

$V_x - R_{sig} I_x = (I_x + g_m I_x R_{sig}) R_L'$

$$V_x = I_x \left(R_{sig}(1 + g_m R_L') + R_L' \right)$$

$$\frac{V_x}{I_x} = R_{sig}(1 + g_m R_L') + R_L'$$

$$\boxed{R_{gd} = R_{sig}(1 + g_m R_L') + R_L'}$$

$$\boxed{R_{db} = R_L'}$$

$$V_x = I_x \left(R_{sig}(1 + g_m R_L') + R_L' \right)$$

$$\frac{V_x}{I_x} = R_{sig}(1 + g_m R_L') + R_L'$$

$$R_{gd} = R_{sig}(1 + g_m R_L') + R_L'$$

$$R_{db} = R_L'$$

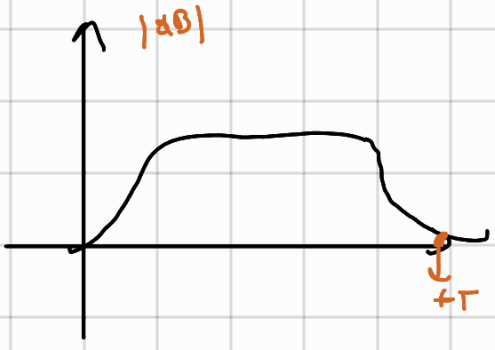
$$Z = RC$$

$$Z_H = R_{gs} C_{gs} + R_{gd} C_{gd} + R_{db} C_{db}$$

$$f_H = \frac{1}{2\pi Z_H}$$

unitygy gain frequency : $f_T = \frac{g_m}{2\pi(C_{gs} + C_{gd})}$

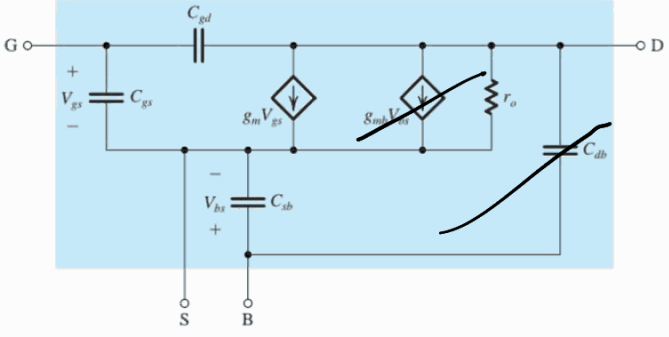
$$f_T = \frac{g_m}{2\pi(C_{gs} + C_{gd})}$$



gain 1 czas nokty

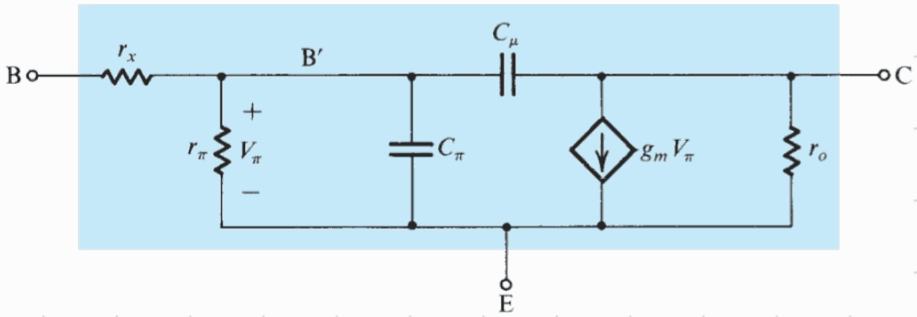
x ile kosztuje nokta

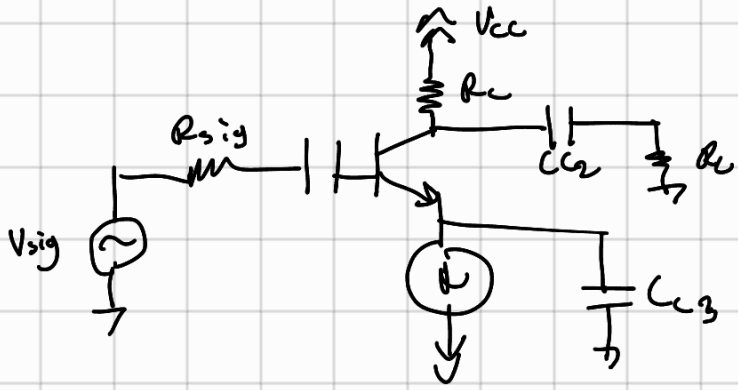
Model

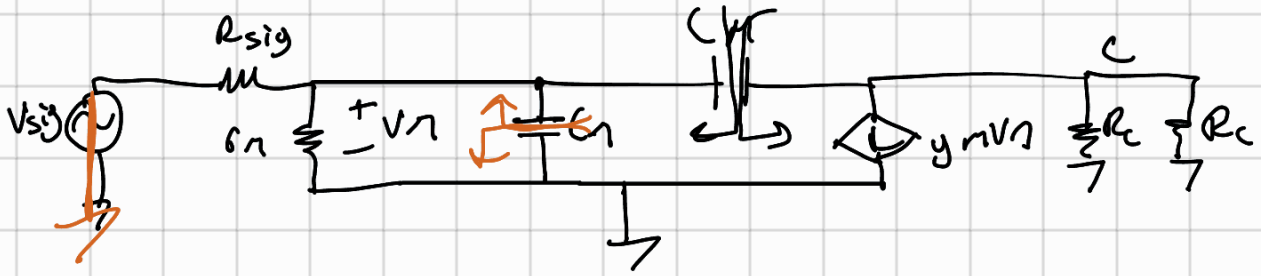


W BJT

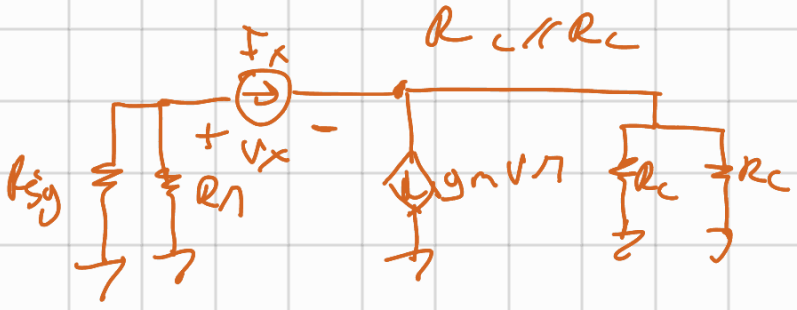
$$C_\mu = \frac{C_{\mu 0}}{\left(1 + \frac{V_{CB}}{V_{0c}}\right)^m}$$







$R_{Cn} = R_{sig}$



Short circuit collector current

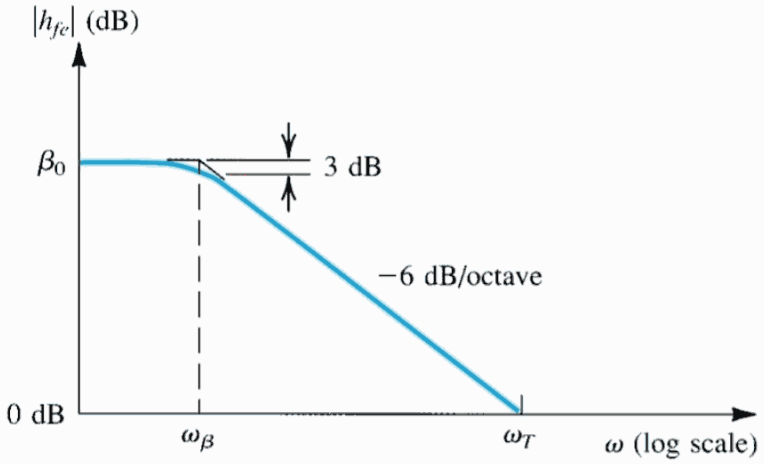
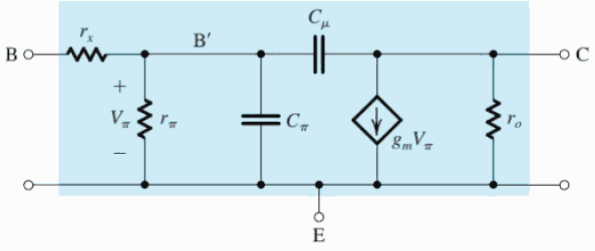


Figure 9.10 Bode for $|h_{fe}|$.

Table 9.2 The BJT High-Frequency Model



$g_m = I_C/V_T$ $\quad r_o = |V_A|/I_C$ $\quad r_\pi = \beta_0/g_m$

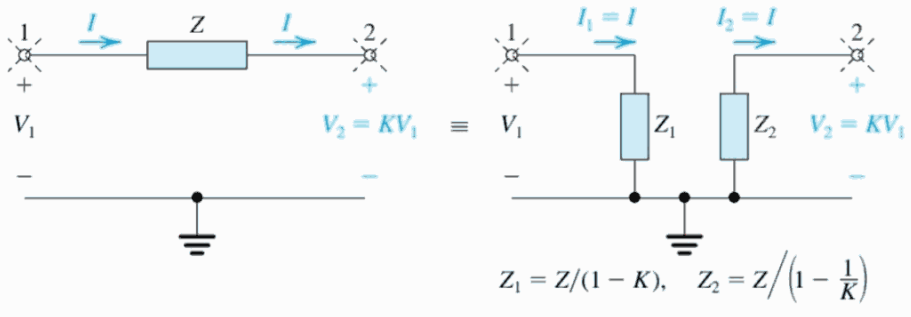
$C_\pi + C_\mu = \dfrac{g_m}{2\pi f_T}$ $\quad C_\pi = C_{de} + C_{je}$ $\quad C_{de} = \tau_F g_m$ $\quad C_{je} \simeq 2C_{je0}$

$C_\mu = C_{jc0}\Big/\left(1 + \dfrac{|V_{CB}|}{V_{0c}}\right)^m$, $\quad m = 0.3\text{–}0.5$

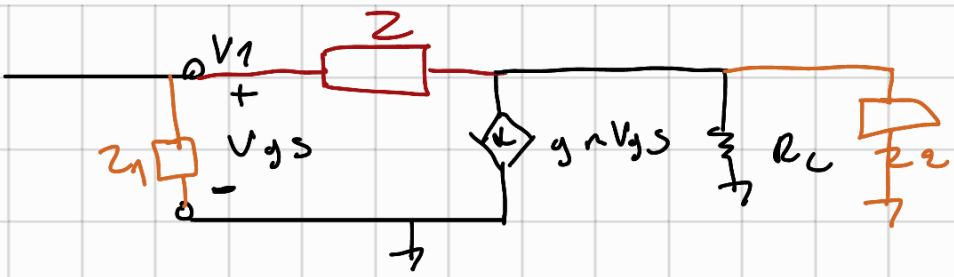
f_H 'da dominant olan en küçük olan

Miller's Theorem

$V_2 = k V_1$



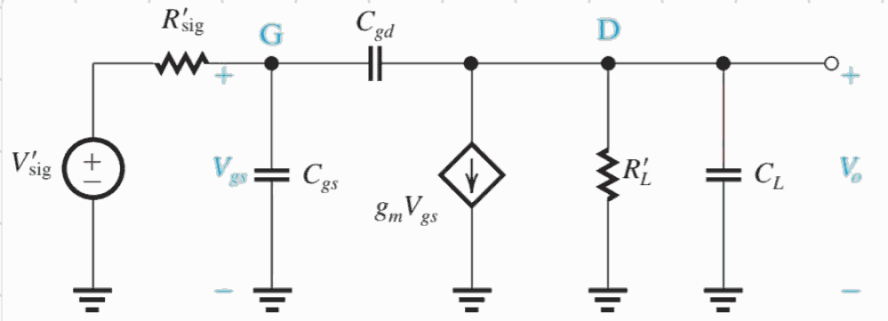
$Z_1 = Z/(1 - K)$, $\quad Z_2 = Z\Big/\left(1 - \dfrac{1}{K}\right)$

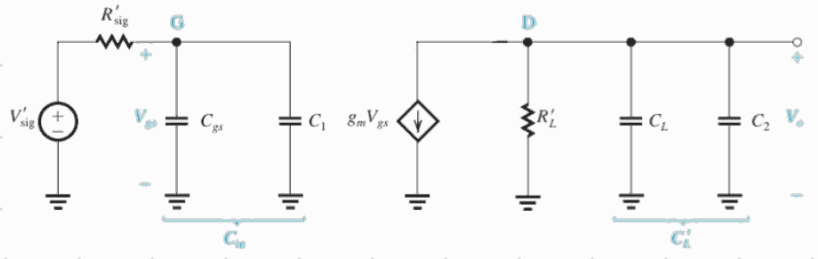


$V_1 = V_{gs}$

$V_2 = -g_m V_{gs} \cdot R_L'$

$|k| \cong g_m R_L'$





Görünen kapasitörler low frekansı belirler

Mosfet: $f_T = g_m / 2\pi (C_{gs} + C_{gd})$

BJT: $f_T = g_m / 2\pi (C_\pi + C_\mu)$

$GB = A_M f_H \qquad f_T = GB$

